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Time series analysis of energy usage patterns by Tanzania large power users

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Abstract

The widespread adoption of smart meters has significantly made available detailed energy consumption data, providing valuable insights into consumer usage patterns. This paper takes advantage of this fact and presents a comprehensive time series analysis of energy usage patterns using smart meters time series data from high-usage power consumption companies in Tanzania. To the best of our knowledge, the study builds upon no any other data-driven research that uniquely focuses on TANESCO smart meter data to draw insights and forecast usage trends. Therefore, a systematic and scientific study is initiated to explore energy usage trends to enhance the power services by understanding customer consumption behavior. Our study introduces robust methods to handle missing data, a prevalent issue in Tanzania smart meter datasets due to power interruptions, ensuring robust energy consumption analysis. It, however, addresses the challenges of inconsistent data availability and significant discrepancies in power usage among different companies in Tanzania by developing methods to work with fragmented data and draw meaningful conclusions about long-term energy trends. The study devises an approach to unify discontinuous datasets for time series studies and analyzes variations in energy consumption patterns to reveal operational and sectoral influences on energy demand, providing insights into the underlying factors driving these variations and their implications for energy management. Our findings suggest that a detailed understanding of temporal consumption patterns can lead to more efficient demand-side management, encompassing improved demand response, power management, power monitoring, and distribution efficiency, resulting in cost-savings and a possibility of enhanced integration of renewable energy sources.

Keywords: Anomaly detection, Clustering analysis, Demand-side management, Energy efficiency, Machine learning, Power utility, Smart metering systems, Time series

Introduction

Countries worldwide are at the center of massive smart meter installations for users on the way toward grid digitization and modernization. These transformations enable the collection of a huge amount of fine-grained smart meter data that can be processed to draw insights, especially now with widely available machine learning and data analytic techniques. The global push toward energy efficiency and sustainability is becoming increasingly critical to understanding consumption patterns, and hence tools that can accomplish this task are very desirable.

This study is particularly interested in deploying time series analysis methodology to analyze data collected over years. Specifically, fine-grained smart meter energy profiles, with data recorded at 20-minute intervals for a select number of large power users in Tanzania from 2017 to 2023, are utilized. Time series techniques are applied to these profiles to derive insights into energy consumption behavior. The deployment of smart metering systems provides access to high-resolution electricity consumption data, offering a wealth of information for understanding energy usage patterns. This fine-grained data enables in-depth insights through various data analytics techniques, as comprehensively reviewed by [1], allowing utility providers and researchers to gain valuable knowledge about consumer behavior and load patterns [2]; [3]. This in turn enables the creation of individual consumer profiles, which can enhance energy management systems by facilitating load forecasting, efficient demand and supply management, and more reliable service [4]; [5].

[6] provides strong evidence that higher-resolution smart meter data significantly improves the detectability of energy consumption trends by offering a more rigorous understanding of patterns, enabling quicker responses to changes, and allowing for the identification of detailed appliance-level consumption behaviors. This is true based on the fact that time variable consecutive consumption values are likely to be closely similar. The claim can as well be validated using statistical measures such as autocorrelation. Small intervals quantify the degree to which past values correlate with future values in a time series; thus, a high autocorrelation coefficient between consecutive data points would support the assertion that near-future consumption is predictable based on recent consumption. Therefore, even though our data is not at the very high resolution discussed in [6], its moderate granularity still allows us to utilize the benefits of time series analysis for meaningful insights and improvements in energy management.

Furthermore, the longevity of the smart meter data, spanning from 2017 to 2023, provides a robust foundation for establishing consumption trends and capturing variations over time. The statistical significance of identified trends can be assessed through regression analysis or time series decomposition. These methods quantify the magnitude and consistency of trends while accounting for seasonality and other factors. Moreover, the data's historical depth allows for the creation of more reliable forecasting models, as longer time series generally improve forecast accuracy.

Traditional electrical and gas meters provided a limited view of energy usage, only measuring total consumption without detailing when that energy was consumed at each metered site [7]. However, with increase in technology and the advent of smart meters, the collection of information about household energy consumption has significantly increased, making it possible for advanced data analysis that was not feasible with traditional meter readings [8]. This enhanced data granularity allows for a deeper understanding of consumption patterns over time, unlike the aggregated data offered by traditional meters [2].

Various studies have utilized smart meter data analytics to improve the management of energy and services, support intelligent grid control, make an accurate forecast, or detect anomalies. [1] provides a comprehensive review of smart meter data analytics in retail markets, encompassing applications such as load forecasting, abnormal detection, consumer segmentation, and demand response, and discuss the latest advancements in

these areas. [9] presents a cluster analysis of Danish household electricity consumption data and demonstrates the importance of pre-processing data to account for temporal components like autocorrelation when using K-Means clustering. This research contributes to the understanding of electricity demand and the development of targeted tariffs by applying data mining techniques and signal analysis methods. Further a work by [10] investigated the problem of inefficient electricity consumption in Tanzanian public institutions, specifically focusing on higher learning institutions (HLIs). The study identified users' behavioral patterns and inadequate technology deployment as the main causes of inefficiency. The findings revealed that a significant majority of participants in the sampled HLIs were not concerned about power-saving strategies while using office facilities. The study proposed interventions targeting both behavioral changes and technological improvements including using smart meters for power saving in HLIs.

In recent years, [11] explored the electricity usage patterns using data mining techniques on smart meter data with a focus on temporal usage behaviors, such as peak usage periods and seasonal variations. [12] and [13] explore the use of ARIMA models for forecasting electricity consumption, demonstrating its potential to compete with existing electricity consumption forecasting techniques. [14] utilizes six time series analysis methods to forecast electricity consumption: simple moving average, weighted moving average, simple exponential smoothing, Holt linear trend, Holt-Winters, and deseasonalization. The monthly electricity consumption data from January 2011 to December 2017 was used to apply these methods and forecast the monthly consumption for 2018, comparing the accuracy of each of these methods using error analysis metrics such as MAE, MAPE, MSE, and RMSE. [15] forecasts smart grid electricity consumption using a bottom-up time series clustering, employing Euclidean and dynamic time warping distances, segments household consumption data derived from AMI. Individual household demand is then predicted using a variety of time series models, including the autoregressive integrated moving average (ARIMA), ARIMA exogenous (ARIMAX), double seasonal Holt-Winters (DSHW), trigonometric, Box-Cox transform, autoregressive moving average errors, trend and seasonal components (TBATS), neural network nonlinear autoregressive (NNAR), and nonlinear autoregressive exogenous (NARX). Results demonstrate superior accuracy (lower MAPE and RMSE) compared to aggregate demand forecasting.

Different researchers have investigated on time series for smart meters with respect to demand time management. For example, [16] explored the use of clustering techniques on 24-hour electricity consumption data from commercial and industrial users. Their aim was to identify common usage patterns across smart meter time series data. By grouping similar energy profiles, they provided insights that could help improve energy efficiency. The study highlights how data-driven analysis can support smarter energy management strategies. [17] explore the use of functional principal component analysis (FPCA) to analyze and predict electricity consumption patterns over time. The method helps identify key variations in energy use by breaking down complex time series data into simpler components. This approach proves effective in capturing both short-term and seasonal usage trends. The study demonstrates FPCA's potential for improving energy demand forecasting and understanding consumer behavior. Also, [18] explored methods for identifying patterns in large-scale electricity consumption data using

time series analysis. They employed distributed computing techniques, applies distributed k-means clustering (through Apache Spark), to efficiently process and cluster vast amounts of energy usage data. The study focused on detecting recurring consumption behaviors in university buildings from 2011 to 2013. Their approach demonstrated the effectiveness of scalable algorithms in uncovering meaningful insights from complex energy datasets.

The goal of clustering time series data is to organize the data into homogeneous groups, maximizing the similarity and dissimilarity within and between groups. This is the first data-driven research that uniquely focuses on TANESCO smart metering data in Tanzania. Therefore, a systematic and scientific study is initiated to explore energy usage trends to enhance the power services through understanding customer consumption behavior. Consequently, enhancements are anticipated in demand response, power management, power monitoring, and distribution efficiency. The study may also provide valuable insights and recommendations on how to detect and capture malicious activities and boost the revenue of the company.

Specifically, the article focuses on:

1. How can time series smart meter data analysis enhance energy management systems?
2. What strategies can be utilized by the stakeholders to enhance energy management systems using smart meter datasets?

In contrast to other analyses, achieving the objectives of this study necessitates addressing the significant challenge of high-frequency missing values in smart meter data. This issue arises from frequent, irregular power cuts, a unique situation within the Tanzanian context. In such a scenario, robust data processing and forecasting techniques are essential to ensure data integrity. Moreover, this research delves into a unique context where it is necessary to tackle the challenge of inconsistent data availability from commercial companies across different years. This requires developing methods to work with fragmented data and draw meaningful conclusions about long-term energy trends. To address the fragmented nature of energy data, an approach is devised to unify discontinuous datasets, making them suitable and useful for time series studies. The data is also characterized by considerable magnitude discrepancies in power usage among different companies in Tanzania. A thorough analysis of these significant variations in energy consumption patterns across diverse companies has the potential to reveal crucial operational and sectoral influences on energy demand. Furthermore, this analysis can offer valuable insights into the underlying factors driving these variations and their implications for effective energy management. This may also provide cumulative insights into energy usage patterns across different geographical locations, sectors, and demographic dynamics in Tanzania.

Our study aligns with this body of work by employing time series analysis to derive insights into energy consumption behavior using fine-grained smart meter data. Time series analysis and decomposition methods [19] are valuable for understanding long-term consumption behaviors and periodic fluctuations in energy data. Techniques like seasonal and trend decomposition using loess (STL) allow for the separation of data

into trend, seasonal, and residual components, providing a deeper understanding of the underlying patterns, see Subsection 3.2. Outlier detection on the residuals can then highlight anomalies or irregularities in consumption.

The rest of the paper is organized as follows: Section 2 presents materials and methods, details the structured methodology employed to process, visualize, and analyze the unique characteristics of the TANESCO smart meter data. This section includes subsections describing the data sources, the process of data acquisition, and the pre-processing strategies adopted to handle discontinuous and fragmented datasets and address data quality and integrity. It also details the specific analytical methods (time series techniques) deployed in the study, as well as the statistical and visualization tools employed to interpret data and the results. Section 3 presents the findings derived from the time series analysis. This section includes subsections showcasing the temporal consumption patterns uncovered and discussing their implications for more efficient demand-side management and potential cost-savings. The paper then proceeds to Section 4, where the limitation of the study is presented and addressed potential constraints and uncertainties in the research. Finally, Section 5 summarizes the key findings of the study and outlines potential directions for future work.

Materials and methods

The methodology employed in this study is structured to address the unique characteristics of the Tanzania smart meter data. Our approach is designed to extract meaningful patterns and trends from the fine-grained energy profiles recorded at 20-minute intervals. As previously highlighted, significant challenges in analyzing these smart meter datasets are the prevalence of missing data due to frequent power interruptions, inconsistent data availability from commercial companies across different years and the fragmented nature of the datasets. Therefore, our methodology incorporates robust methods to handle these missing values, ensuring a reliable energy consumption analysis. Furthermore, recognizing the inconsistent data availability from commercial companies across different years and the fragmented nature of the datasets, our methodology includes strategies to unify nonuniform discontinuous datasets, making them suitable for comprehensive time series studies. The subsequent subsections outline the data sources, data acquisition, pre-processing strategies, and the analytical methods deployed in this study to derive insights into energy usage patterns.

Data acquisition

The study uses secondary data from smart meter readings given by Tanzania Electric Supply Company (TANESCO) that is accessed through its website [20]. The raw data for this study was acquired by downloading numerous Excel files from this website. Each file typically represented a daily report and contained smart meter readings. These reports featured data for one or more specific customers for that particular date, identified by details such as Customer Name and Meter Number. It was observed that each Excel file for a specific date could contain data for a varying number of companies. There was no fixed number of companies present in each daily report, with some files including data for fewer companies and others for more.

Initially, the Excel files featuring electric power consumption in kilowatt-hours (kWh) collected by AMR at 20-minute intervals throughout 24 hours were downloaded for the period spanning from January 1, 2017 to December 31, 2023. Upon examining the downloaded data, it was apparent that many customers had highly unavailable or missing data across the years. To ensure a robust time series analysis, the companies are flagged with significant inconsistencies in data availability. Our selection process focused on identifying companies that had data throughout the majority of the study period. After this initial assessment and flagging of numerous customers with incomplete datasets, eight companies are retained that demonstrated relatively consistent data availability across the years. For anonymity, these eight companies were subsequently referred to as Company A, B, C, D, E, F, G, and H. The electric power consumption data for these selected companies included measurements for kWh (kilowatt-hours) and kvarh (kilovar-hours), recorded at 20-minute intervals, making a total of 184032 rows of datasets, having separate column values for Import and Export

Data pre-processing

Data pre-processing was a critical stage involving the preparation of raw datasets for subsequent analysis by addressing inconsistencies, invalid entries, and missing values. The initial step involved downloading numerous highly unstructured raw data Excel files, each typically representing a daily report for a number of customers, highlighting the crude and high-volume nature of the fine-grained smart meter data cluttered within files metadata. Each Excel file corresponded to a particular date and contained energy consumption data for one or more specific customers. These reports included details such as the Customer Name, CT Ratio, VT Ratio, and the Meter Number. The energy consumption data, including kWh (kilowatt-hours) and kvarh (kilovar-hours), was recorded at 20-minute intervals throughout the 24-hour period. For each 20-minute interval, the data was split into measurements for both Import and Export. Additionally, each daily report contained a summary section providing the total, maximum, and minimum values for Import kWh, Export kWh, Import kvarh, and Export kvarh for that specific day. These files were timestamped and provided a chronological record of the energy usage for each of the specified customers on a daily basis.

This necessitated optimized programming strategies using Python Pandas and NumPy arrays to manage the computational intensity associated with processing these numerous files, effectively reducing computational overhead and enabling timely data cleaning and processing. The pre-processing steps implemented included the removal of invalid timestamps using conditional filters. Data corresponding to the year 2024 was excluded due to its limited scope for meaningful generalization. To improve data accessibility, column renaming and restructuring were performed. Columns with substantial lengths of zeros or those with zero-sum values, deemed noncontributory, were identified and excluded. Missing values, a prevalent issue in Tanzanian smart meter datasets caused by power interruptions, were addressed using a forward-filling imputation strategy. To account for temporal granularity and facilitate further analysis, the data were resampled into daily and monthly aggregates using the resample function with summation, ensuring accurate representation for temporal and statistical analysis. It is important to note that a notable challenge encountered, involved addressing inconsistencies and structural

variations within the datasets. These variations included changes in labeling conventions, alterations in data structures, and instances where single columns were split into multiple columns, or similar data types were recorded across differently labeled columns. While the implemented pre-processing measures such as standardization of column names and data structures aim to maintain consistency, the inherent irregularities of this real-world smart meter data underscore the critical importance of rigorous data cleaning and standardization to ensure the accuracy and reliability of subsequent analysis. The transformation process resulted in a hierarchical, chronological format, where rows featured chronologically ordered timestamps and represented KWH Import consumption, and columns represented customer names, ultimately transforming the data into a structure suitable for time series analysis.

The learning effect of a prediction model is often affected by the quality of the input data; therefore, appropriate processing of the data is key to improving the accuracy of the prediction model.

Exploratory data analysis (EDA)

EDA encompassed statistical summarization and visualization to uncover patterns, trends, and anomalies. Descriptive statistics offered insights into data distribution, central tendencies, and variability. Summary tables and histograms helped in identifying the distribution of consumption, revealing whether the data followed a normal distribution or exhibited skewness due to high outliers. The calculation of interquartile ranges (IQR) facilitated initial outlier identification, offering a foundation for deeper anomaly detection. Visualizations such as line plots and boxplots revealed seasonal trends, cyclic behavior, and outliers across different companies and time periods. Line plots of time series data were employed to observe trends over time, highlighting both seasonal and cyclical behaviors in electricity consumption. Seasonality was evident in repetitive patterns occurring at regular intervals, such as daily or monthly electricity usage cycles. Boxplots were also used to display the spread of data across companies, illustrating intercompany variations and highlighting outliers.

Structural time series analysis model

This study employs structural time series decomposition to dissect electricity consumption data into its fundamental components: trend, seasonality, and irregular components. Unlike purely data-driven methods like seasonal and trend decomposition using loess (STL) [8], a structural time series approach allows for explicit modeling of these components, enabling a more nuanced understanding of underlying consumption patterns. Specifically, state-space models are utilized, which facilitate the dynamic estimation of trend and seasonal components, thereby capturing evolving consumption behaviors and periodic cycles. The irregular component, representing unexplained variations, is subjected to rigorous outlier detection. Instead of relying solely on standard deviations, statistical tests are incorporated for normality and examined the distribution of residuals for potential deviations from expected behavior. This approach ensures that identified outliers are not merely extreme values but represent genuine anomalies that deviate significantly from the model's predictions.

Structural time series models, formulated within a state-space framework, offer a robust approach to decompose a time series $X(t)$ into its constituent parts: trend $T(t)$, seasonal component $S(t)$, and irregular component $R(t)$. This decomposition is expressed using an additive model [21]:

$$X(t) = T(t) + S(t) + R(t). \quad (1)$$

Unlike traditional seasonal decomposition methods, such as STL, structural time series models in (1) allow for the dynamic estimation of trend and seasonal components, accommodating time-varying patterns.

In some cases, a multiplicative model is more appropriate, especially when the seasonal variations increase with the level of the time series [21, 22]:

$$X(t) = T(t) \cdot S(t) \cdot R(t). \quad (2)$$

An alternative approach (2) (provided that $X(t) > 0$ for all t) is to apply the linear models on the multiplicative data, that is,

$$\log(X(t)) = \log(T(t)) + \log(S(t)) + \log(R(t)). \quad (3)$$

Outlier detection is performed on the irregular component $R(t)$, which encapsulates deviations not explained by the trend or seasonal patterns. This study begin by examining the distribution of $R(t)$ using statistical tests for normality, such as the Shapiro–Wilk test, to ensure the validity of subsequent outlier detection procedures. If normality assumptions are violated, alternative distributions or robust estimation methods are considered. For normally distributed residuals, outliers are identified based on their standardized deviations from the mean [21]. Specifically, a residual $R(t)$ is flagged as an outlier if:

$$|R(t) - \mu_R| > \tau \cdot \sigma_R(t) \quad (4)$$

where μ_R is the mean of the residuals, σ_R is the standard deviation, and τ is a threshold parameter. In this study, τ is set to 3, aligned with the empirical rule for normally distributed data.

Following outlier identification, the flagged observed values $X(t)$ are compared with their expected values $T(t) + S(t)$. This comparison provides a clear visualization of the magnitude and direction of anomalies, facilitating domain-specific interpretation and further analysis. For instance, large positive deviations might indicate operational interruptions or reporting errors, while negative deviations could signal periods of reduced consumption.

Clustering analysis

Clustering is a crucial unsupervised learning technique for segmenting data based on shared characteristics [23]. To prepare for clustering, data standardization was achieved using StandardScaler, ensuring that features had comparable scales.

The Elbow method is a widely used technique to determine the optimal number of clusters in data segmentation, such as analyzing the power consumption of eight high-use companies in Tanzania [24]. It involves plotting the within-cluster sum of squares (WCSS) against the number of clusters and identifying the elbow point where the rate

of decrease in WCSS slows significantly. For the companies in question, the method revealed an optimal cluster count around a value of 0.2 for the WCSS as seen in Fig. 1.

Observing the graph line, inertia drops sharply from 1 to 3 clusters, indicating that increasing the number of clusters significantly improves data separation. Beyond 4 clusters, the rate of decrease slows considerably, suggesting that would lead to only minor improvements while increasing computational cost and potential overfitting. Using the Elbow method, stakeholders can focus on significant consumption trends while avoiding overfitting the data with too many clusters, making it a practical approach for energy management in Tanzania's context. Silhouette scores, key tools for optimal cluster determination, were also employed to assess and validate the quality of clusters [24].

K-means clustering of eight companies based on power consumption helps segment the companies into groups with similar energy usage patterns. By applying the K-means algorithm, the data is divided into distinct clusters, allowing for targeted analysis of consumption behaviors. The K-means algorithm aims to partition n observations into k clusters, where each observation belongs to the cluster with the nearest mean (cluster center or centroid), serving as a prototype of the cluster. Mathematically, the objective is to minimize the within-cluster sum of squared distances (WCSS) [25]:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

where J is the WCSS, k is the number of clusters, C_i is the i -th cluster, x represents a data point, μ_i is the centroid of the i -th cluster and $\|x - \mu_i\|$ is the Euclidean distance between data point x and centroid μ_i .

This iterative process assigns each data point to the nearest cluster and recalculates the centroids until convergence is achieved. To enhance the clustering process, principal

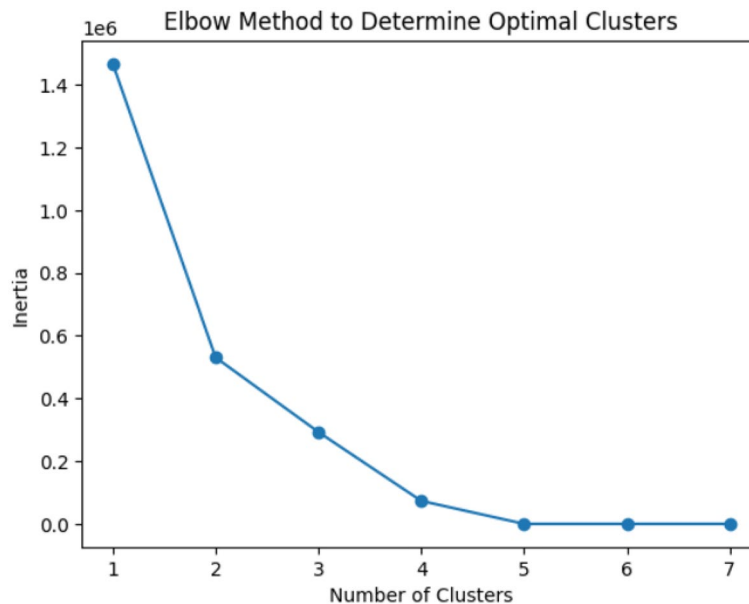


Fig. 1 Clustering Analysis using Elbow method. In this figure, the x -axis represents the number of clusters, while the y -axis shows inertia, which measures how tightly data points are grouped within clusters

component analysis (PCA) is used to reduce the dimensionality of the data, capturing the most significant components that explain the variance in power consumption. PCA simplifies the data by transforming it into a smaller set of uncorrelated components while retaining essential information. This reduction in dimensionality helps avoid overfitting and improves the efficiency of the K-means algorithm, as indicated in 2 [26]. PCA achieves dimensionality reduction by finding orthogonal linear combinations of the original variables (principal components) that capture the maximum variance. The eigenvectors are sorted by their corresponding eigenvalues, and only the top k eigenvectors (corresponding to the largest eigenvalues) are retained, thus reducing the dimensionality [27]. K-means clustering segmented companies into 2 distinct groups based on consumption patterns. Scatter plots, Fig. 2, depicted clear cluster separations, highlighting unique consumption behaviors among companies. Additionally, quantile-based thresholds classified companies into low, medium, and high-consumption groups to provide more insights on the natures of companies and their consumptions.

Results and discussion

Detailed understanding of temporal consumption patterns derived from smart meter data can lead to more efficient demand-side management and cost-savings. It can have a far reaching effect to promoting demand-side management strategies, such as shifting nonessential operations to off-peak (less busy) hours, optimizing energy utilization and enhanced energy distribution efficiency while supporting unique and priority sectors such as industrial growth. The study is expected to provoke many related future works aiming to advance the understanding and management of energy consumption in the Tanzania context.

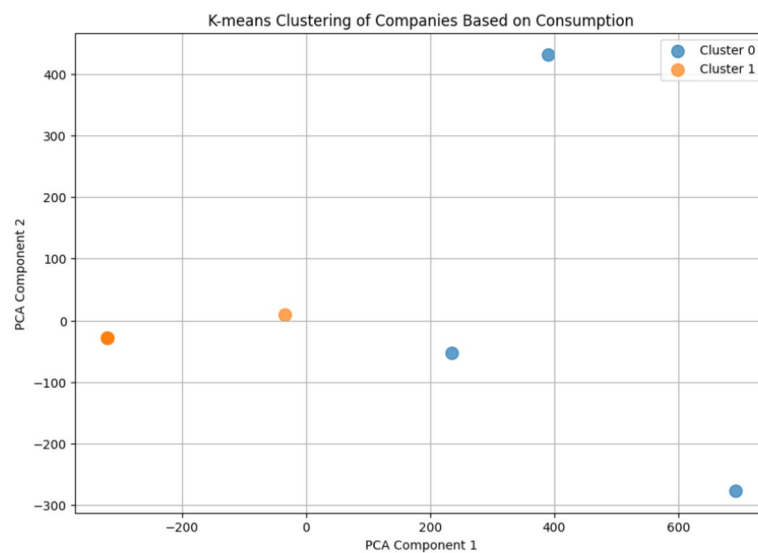


Fig. 2 This scatter plot illustrates the results of K-means clustering applied to companies based on their electricity consumption. The data points are plotted along two principal component axes (PCA Component 1 and PCA Component 2) extracted from the original high-dimensional consumption data that was reduced to two dimensions for visualization. Specifically, Cluster 0 appears to have a wider spread along PCA Component 1, indicating greater variability in electricity consumption patterns within this group, whereas Cluster 1 remains relatively compact

Analysis of power consumption patterns

Our results align with our objectives of understanding consumer behavior and identifying temporal trends in energy usage. Analyzing power consumption patterns for six high-usage companies in Tanzania revealed several notable trends and insights that highlight the dynamics of energy demand in the region. These companies, spanning industries such as manufacturing, mining, and commercial services, exhibited distinct consumption behaviors influenced by their operational schedules, production demands, and external factors like seasonal variations from 2017 to 2024, as seen in Fig. 3. The data also revealed patterns of inefficiency, such as sharp spikes in off-peak hours, likely caused by equipment maintenance or standby operations. Additionally, comparisons across the companies highlighted the pattern similarity of power consumption usage throughout the time coverage except for few.

Companies A, C, F, and H have large power consumption compared to B, D, E, and G; however, all were categorized as large power users companies. During the analysis, it was found that some companies do not have much data; hence, it is difficult to judge whether the smart meter was not sending data or if there was abnormal behavior in the data collection. For example, the company F experienced very low power consumption from the year 2022 onwards compared to previous years, which may also raise abnormal behavior. This is similar to the study by [28] that analyzed the power consumption using smart meter data in data mining that revealing how the user behaviors and appliance usage patterns can be segmented for better energy profiling. Ultimately, understanding consumption behavior is essential for promoting sustainable and efficient energy use [16]. Also, by analyzing these patterns, utilities and consumers can detect inefficiencies, forecast demand, and implement energy-saving strategies. It supports load management, billing optimization, and infrastructure planning.

The findings underscore the need for targeted energy efficiency programs, particularly for the industrial sectors, which could significantly reduce operational costs and grid strain. Moreover, the analysis suggests that promoting demand-side management strategies, such as shifting nonessential operations to off-peak hours, could enhance energy efficiency. These insights provide a valuable foundation for stakeholders in Tanzania to implement informed policies and practices to optimize energy utilization while supporting industrial growth. Summing up, the significant variations in energy consumption patterns among the eight companies (Fig. 3) demonstrates the need for targeted energy efficiency programs tailored to different industrial sectors, which aligns with our study's objective of understanding consumer behavior.

Predictable patterns for energy usage (Trends, Seasonal, and Residual)

The decomposition plot of high-power consumption decomposition graph provides critical insight into the underlying patterns that influence energy consumption. Breaking down the time series data into trend, seasonal, and residual components highlights key drivers of consumption. The trend component reveals a gradual increase in power demand over time, reflecting the growth of industrial activities and economic development in the region. Seasonal patterns indicate consistent peaks during the dry season, likely due to increased mining activities and cooling demands,

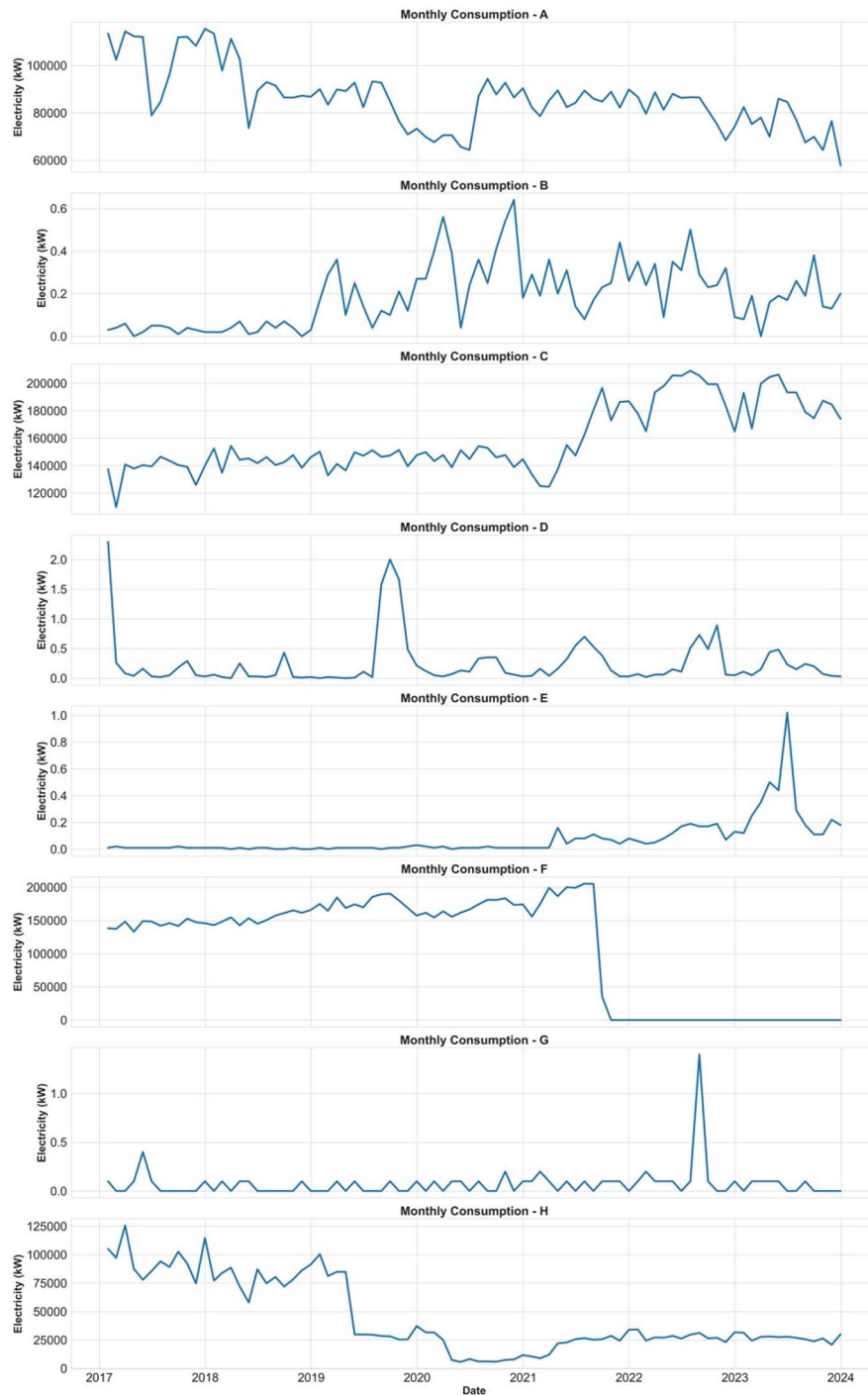


Fig. 3 Consumption profiles for each company

while dips occur during the rainy season when energy usage is relatively lower. Figures 4, 5, 6, 7, 8, 9, 10, and 11 describe the company's trends, seasonal, and residual components for each company. The weekly cycles show increased consumption during the weekdays, aligning with the operational schedules of industries and

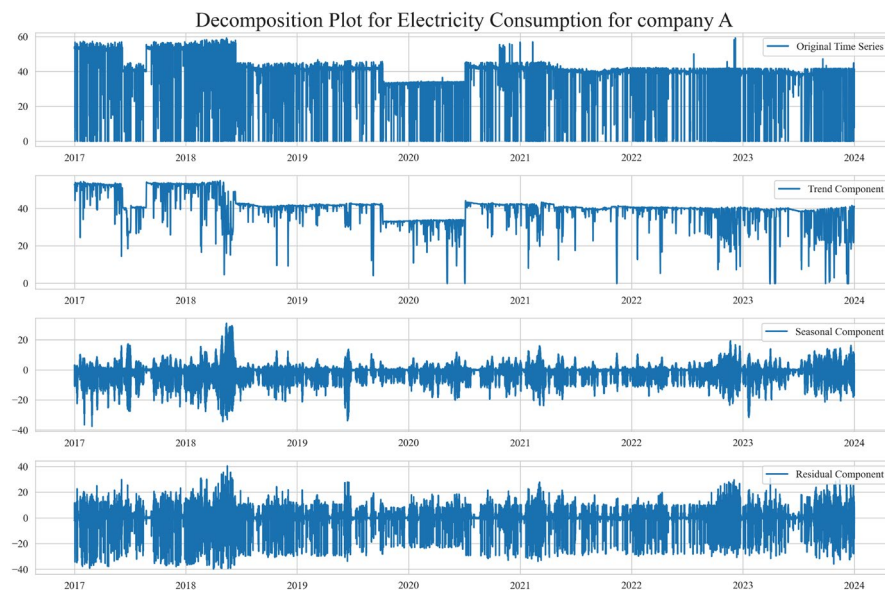


Fig. 4 Trends, Seasonal, and Residual for Company A

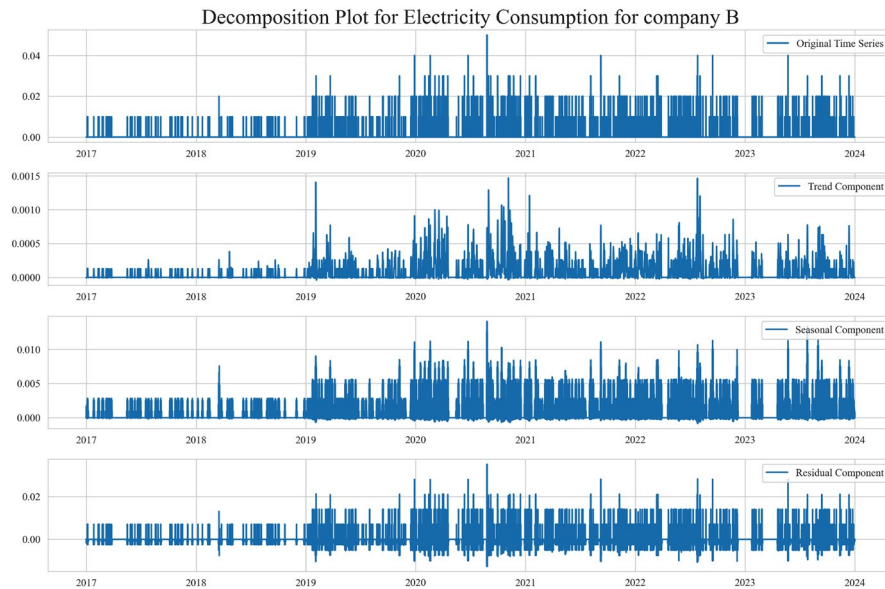


Fig. 5 Trends, Seasonal, and Residual for Company B

commercial entities. The residual component captures anomalies and irregularities, such as unexpected spikes or drops, often associated with power outages or equipment failures. These findings emphasize the importance of understanding the dynamics of temporal consumption for effective energy planning. The decomposition plot not only helps identify predictable patterns, but also uncovers inefficiencies and irregularities that can guide targeted interventions to optimize power use in Tanzania's high-consumption sectors.

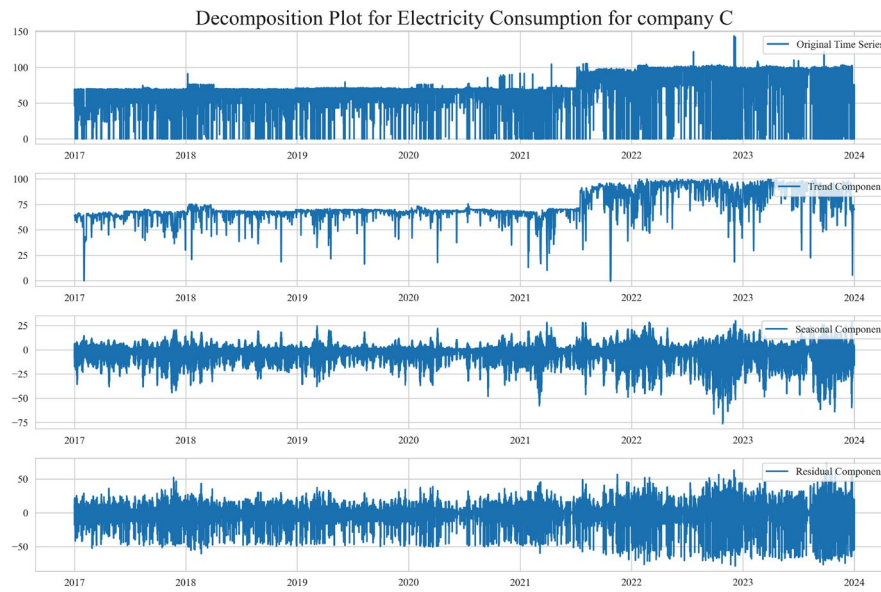


Fig. 6 Trends, Seasonal, and Residual for Company C

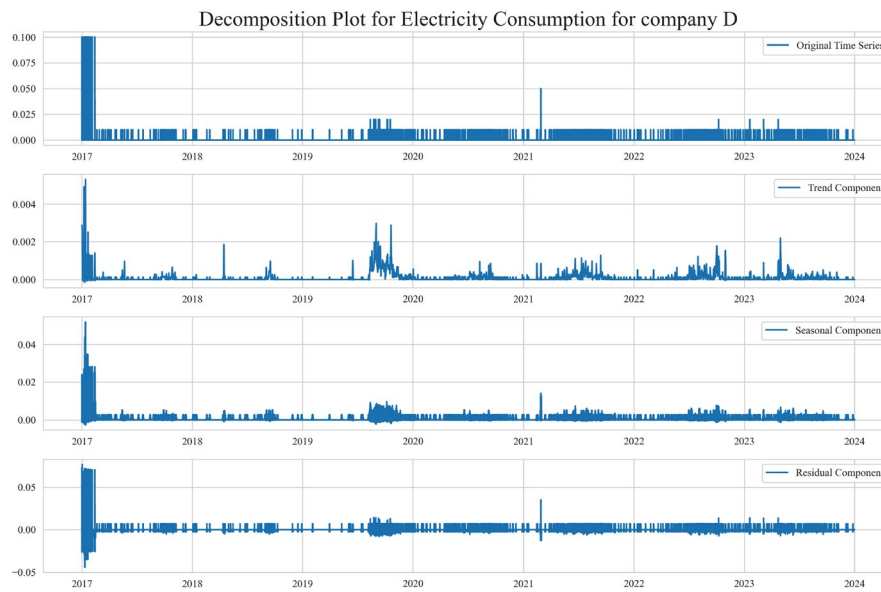


Fig. 7 Trends, Seasonal, and Residual for Company D

The decomposition plot of the high-power consumption decomposition graph analyzes data into trend, seasonal, and residual components. Figures 4, 5, 6, 7, 8, 9, 10, and 11 describe the company's trends, seasonal, and residual. The original time series for Company A's power consumption reveals frequent spikes and dips. The power consumption data contains all trends and seasonality characteristics, which can be useful for forecasting. The seasonal patterns observed in Company A's energy consumption (see Fig. 4) support the effectiveness of the structural time series decomposition method employed for identifying cyclical variations. The model successfully isolated the seasonal

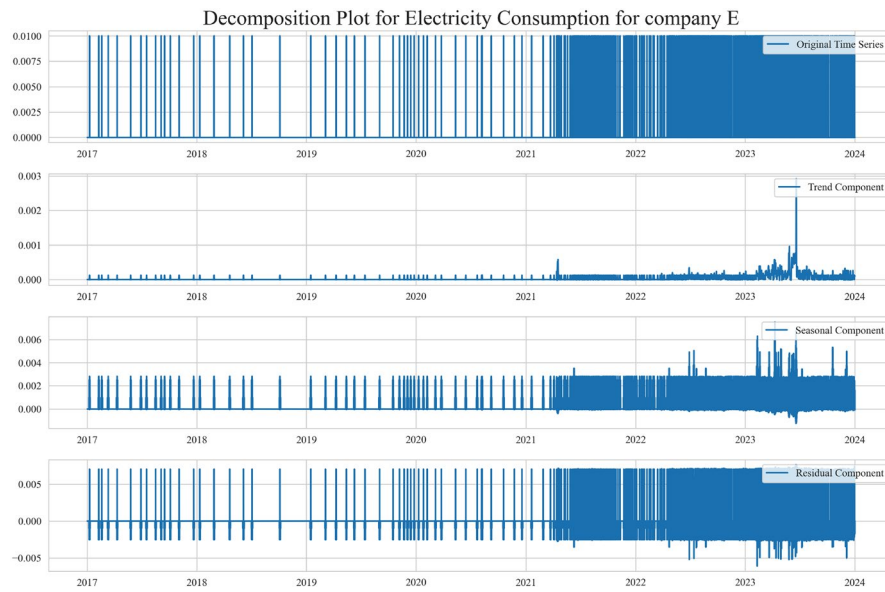


Fig. 8 Trends, Seasonal, and Residual for Company E

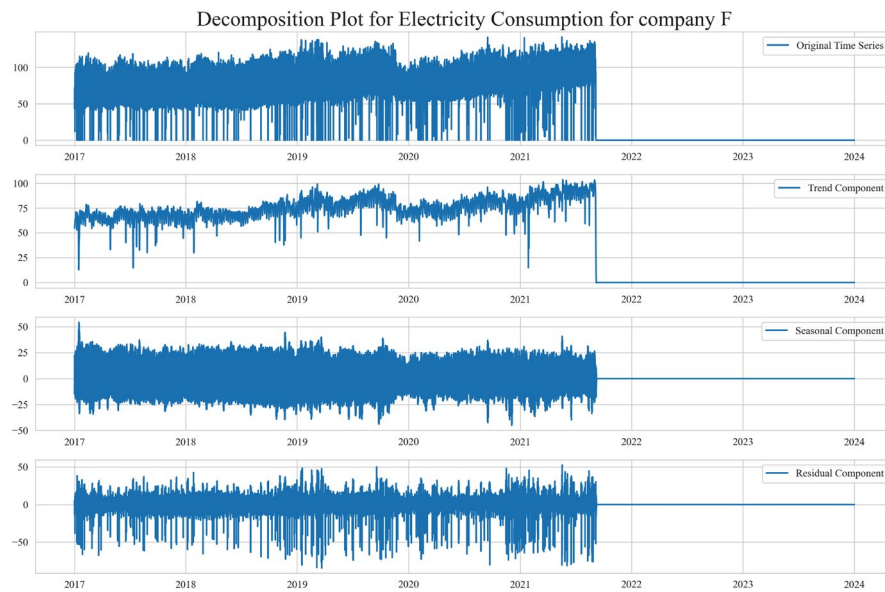


Fig. 9 Trends, Seasonal, and Residual for Company F

component from the trend and residual components. With the duration from 2017 to 2023, the utility company can manage to control and monitor power usage. This has also been depicted by [12] and [29], who did power forecasting using time series data.

Overall, Company A's consumption appears more erratic and less seasonally consistent, potentially reflecting unstable operations or external disruptions. Company B's decomposition highlights a significant structural change in electricity usage around 2019, driven by both trend growth and evolving seasonal behavior. For Company C, it exhibits consistently high consumption with a clear upward trend and strong seasonal

irregular events or external factors influencing consumption.

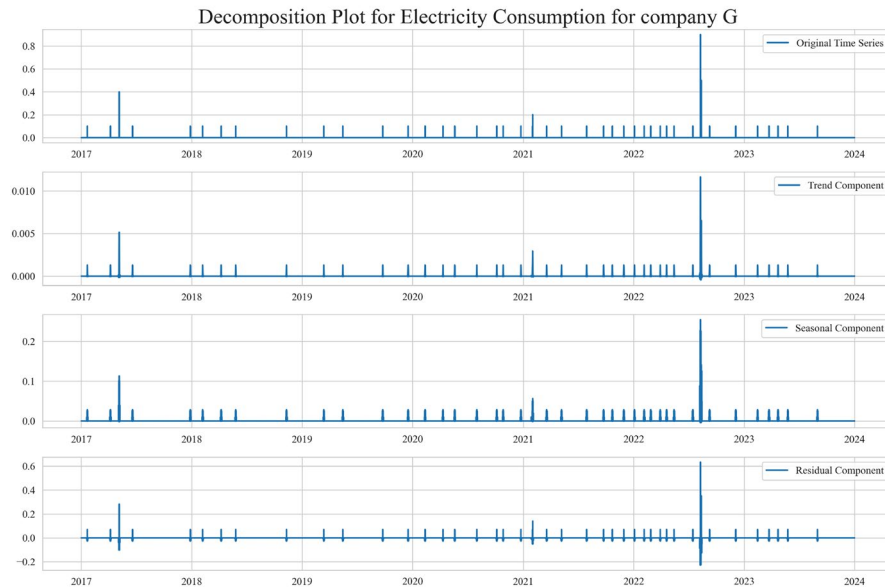


Fig. 10 Trends, Seasonal, and Residual for Company G

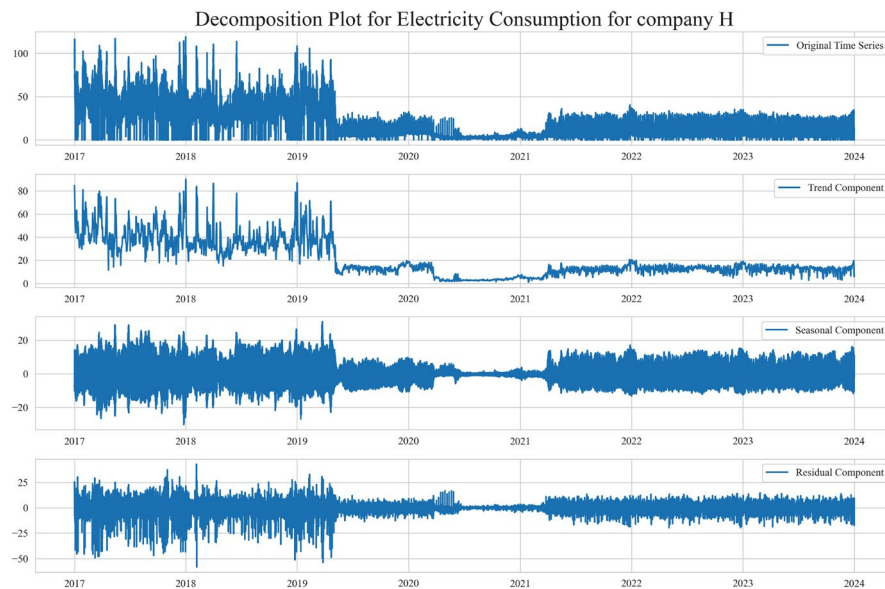


Fig. 11 Trends, Seasonal, and Residual for Company H

fluctuations throughout the observed period, although it also displays large residual variations indicating possible anomalies or irregular consumption behavior. In contrast, Company D's electricity use is minimal and highly sporadic, concentrated mainly at the beginning of the period with a short-lived trend and limited seasonal structure, suggesting inconsistent operations or infrequent power use. Company E shows highly sparse and erratic original time series data, especially before 2021, with many periods of zero or near-zero values, indicating potential data collection issues or low consumption.

Company F presents a much more stable and continuous time series from 2017 to early 2021, with a clear and smooth trend showing moderate growth. Company G has minimal or highly irregular electricity demands with very limited or sporadic activity, while Company H demonstrates substantial operational change or downsizing.

Several studies have discussed also the decomposition of energy consumption into trend, seasonal, and residual components for an energy management system. Studies by [30] and [31] explore these decomposition methods for energy time series data to improve forecast accuracy. In their work, they emphasized the importance of separating seasonal demand (e.g., air conditioning in summer) from long-term growth trends. [32] applied double seasonal exponential smoothing models to capture both weekly and yearly seasonality in electricity demand. Another study by [33] utilized high-resolution smart meter data to analyze household-level load patterns, revealing distinct seasonal effects and irregular events.

The nature of other Figs. 3 to 9 shows similar trends of growing energy usage. The main difference seen in Company B in Figure 5 is that it started at a much lower level and has grown significantly over time compared to Company A. Company C in Fig. 6 has a more stable electricity usage pattern with fewer extreme variations than Companies A and B, indicating a more predictable and controlled consumption behavior. With regard to Company D of Fig. 7, the residual component, which captures irregularities, appears relatively stable with minor deviations, indicating that the trend and seasonal components explain most variations.

The original time series for Company E in Fig. 8 shows a sharp increase in activity starting around 2021. The trend component indicates a gradual upward movement that accelerates in 2022 and 2023, possibly due to increased operational activity or expansion. The seasonal component demonstrates periodic fluctuations, consistent throughout the observed period but becoming more noticeable post-2021. The residual component shows high variability and irregular spikes, especially after 2021, indicating potential anomalies or external factors influencing consumption.

The decomposition plot for Company F's in Fig. 9 highlights consistent and relatively stable patterns over time. The original time series shows fluctuations with periodic peaks and troughs, suggesting a regular consumption cycle. The trend component indicates a slight upward trend, reflecting gradual growth in electricity usage. The seasonal component captures repetitive patterns that persist throughout the observed period, implying predictable cyclical behavior. The residual component exhibits random noise and occasional spikes, suggesting intermittent, irregular events or external factors influencing consumption.

The decomposition plot for Company G's in Fig. 10 reveals a sparse and irregular usage pattern, with occasional spikes in energy consumption. The trend component remains near zero, suggesting minimal long-term growth or decline. The seasonal component shows small periodic fluctuations, but overall consumption is infrequent. The residual component reflects sharp, irregular peaks, indicating unpredictable energy usage, likely due to sporadic or intermittent activities.

The decomposition plot for Company H's in Fig. 11 shows a decline in the trend component, indicating a steady reduction in energy usage since 2019. The seasonal component remains relatively stable, suggesting recurring consumption cycles, while the

residual component shows high volatility early on but stabilizes in later years. This pattern suggests that Company *H* either reduced operations, improved energy efficiency, or experienced external disruptions, leading to lower electricity consumption.

Distribution and dynamics of time series data

The distribution and dynamics of time series data can be clearly described by using the boxplot method. The boxplot method is a valuable tool in time series analysis, primarily for understanding the distribution, variability, and outliers in data at different time intervals.

Figure 12 illustrates the distribution of electricity consumption across eight companies (*A* to *H*), revealing differences in median consumption, variability, and outliers. Companies *C* and *F* exhibit the highest median electricity usage, suggesting they are among the largest consumers. In contrast, Companies *B*, *D*, *E*, and *G* display almost no variation, indicating minimal consumption compared to the rest. Company *H*, while having a lower median than *C* and *F*, still shows a wide spread, implying fluctuations in its electricity usage. Company *A*, with a relatively smaller interquartile range (IQR), suggests more stable but slightly varied consumption. Outliers are prominent in Companies *C*, *F*, and *H*, where extreme values above the upper whisker indicate occasional spikes in electricity consumption. These variations suggest that these companies experience periods of significantly higher energy usage. Company *G*, despite its low overall consumption, has some outliers, suggesting occasional fluctuations. On the other hand, Companies *B*, *D*, and *E* exhibit minimal, indicating stable consumption patterns. The variability in power consumption suggests opportunities for targeted interventions, such as energy audits or efficiency programs. The BoxPlot analysis can thus provide actionable insights for optimizing energy allocation and policy making tailored to the diverse industrial landscape of Tanzania.

Several studies have explored the distribution and dynamics of time series data with a focus on visualization and analysis techniques. For instance, [34] discuss how a box plot

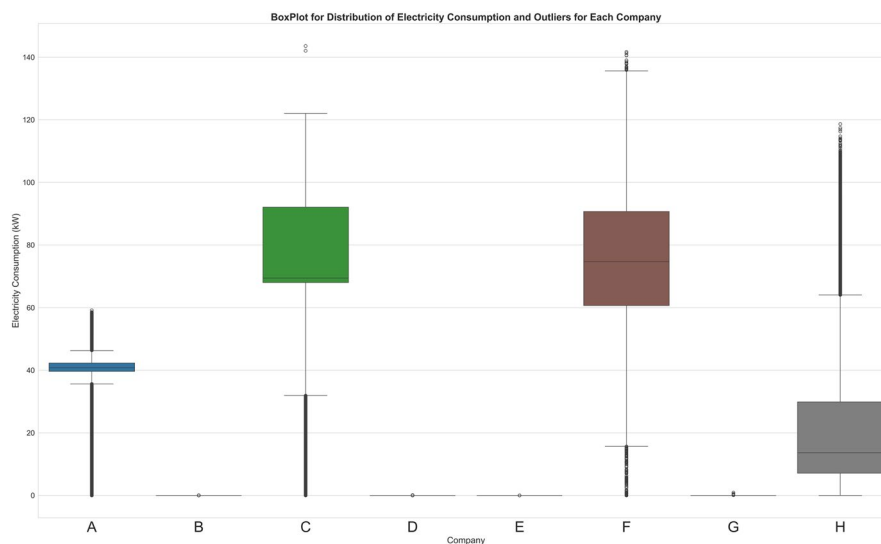


Fig. 12 Distribution of Electricity Consumption for Each Company

can reveal underlying patterns in time series data before formal modeling. [22] emphasize understanding temporal dependencies such as autocorrelation and seasonality to capture dynamic behavior. [31] utilize box plots to detect outliers and changes in distribution across time windows in energy consumption data. The work by [35] integrates visualization tools with forecasting techniques to enhance interpretation of time series distribution and dynamics. Lastly, [36] demonstrate how changes in distributional properties over time can indicate regime shifts or structural breaks in financial time series.

Moreover, the analysis of value deviation is crucial for identifying periods where model accuracy drops, enabling targeted model refinement, anomaly detection, and operational audits to better understand and manage electric power systems. The graphs for Companies A through H depict time series of the standard deviation between actual and expected electricity consumption values, effectively showing deviations or anomalies in power usage over time. In Fig. 13 Company A exhibits relatively consistent behavior with occasional spikes, indicating mostly stable operations with intermittent anomalies. In Fig. 14, Company B shows a similar pattern but with slightly higher variability, suggesting more frequent deviations, possibly due to operational irregularities or external disruptions. Company C in Figure 15 maintains a low standard deviation across the timeline, reflecting high predictability and stable consumption patterns. Company D, as seen in Fig. 16, demonstrates sporadic yet significant deviations, which may indicate irregular operations or data inconsistencies. Company E in Fig. 17 displays a noticeable shift starting around 2021, with a dense cluster of high deviations suggesting a change in operational dynamics or data reporting practices. Company F has a relatively stable trend with moderate fluctuation, suggesting consistent consumption with periodic adjustments as shown in Fig. 18. Company G presents isolated yet very high deviations, implying rare but substantial anomalies in power consumption, which could be outliers or major one-off events as described in Fig. 19. Lastly, in Fig. 20 Company H shows a dramatic decline in standard deviation over time, possibly reflecting improved forecasting, stabilization of operations, or a reduction in power usage variability. Collectively, these plots reveal distinct operational profiles, with some companies exhibiting high predictability and others experiencing notable variability or transitions in consumption behavior. The absence of strong deviation patterns in some periods could suggest times of stable and predictable consumption, where the forecasting model aligns well with actual usage. Furthermore, the clustering of high deviation values in the rest of the companies in certain periods (e.g., during or after 2020) may hint at structural or behavioral shifts-perhaps due to the COVID-19 pandemic, regulatory changes, or economic disruptions. These could have altered energy usage patterns, rendering earlier predictive models less effective.

Limitations of the study

Despite employing a forward-filling imputation strategy to address missing data resulting from power interruptions, and developing methods to unify discontinuous datasets arising from inconsistent data availability, the inherent estimations in these processes introduce a degree of uncertainty. Furthermore, while our analysis accounts for significant variations in energy consumption patterns, the high data fluctuation and variability, potentially influenced by frequent power cuts, may impact the definitive capture of subtle underlying trends. Nonetheless, these crucial steps in handling missing and

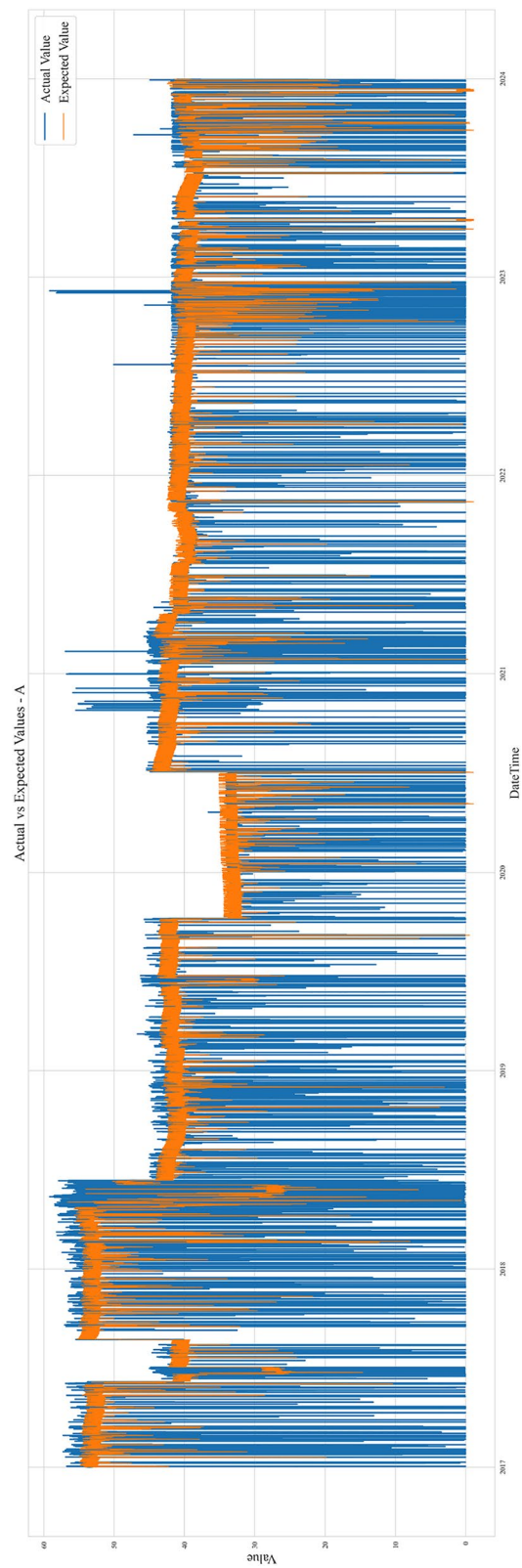


Fig. 13 Analysis of value deviation in Company A

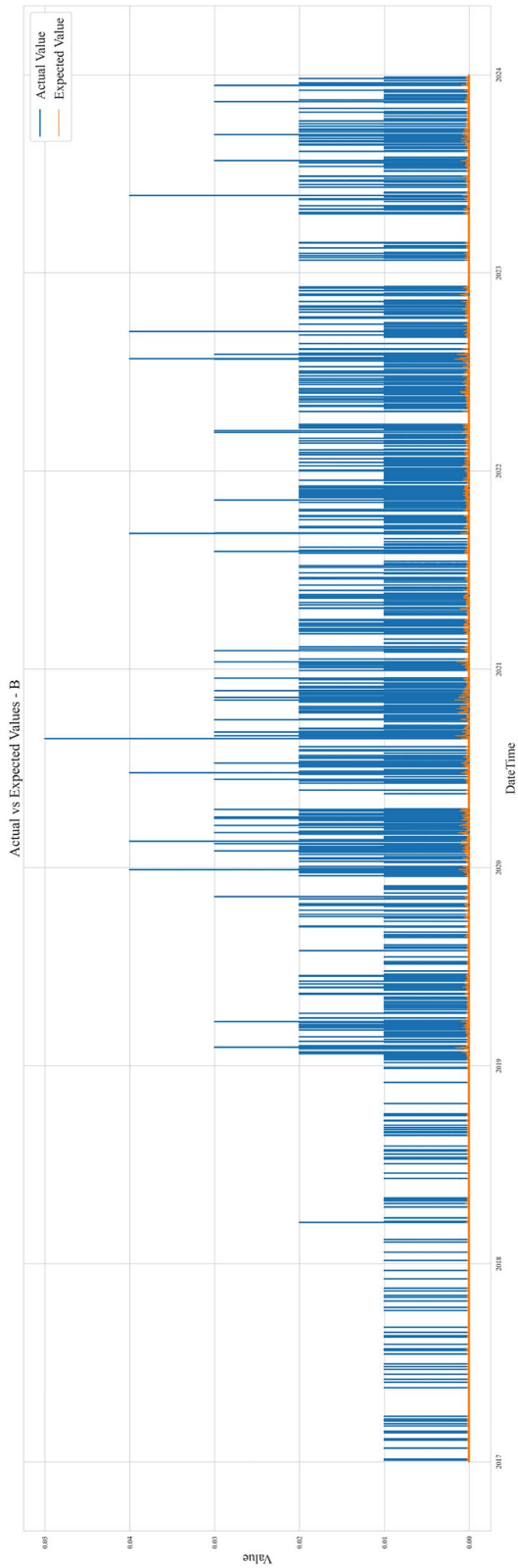


Fig. 14 Analysis of value deviation in Company B

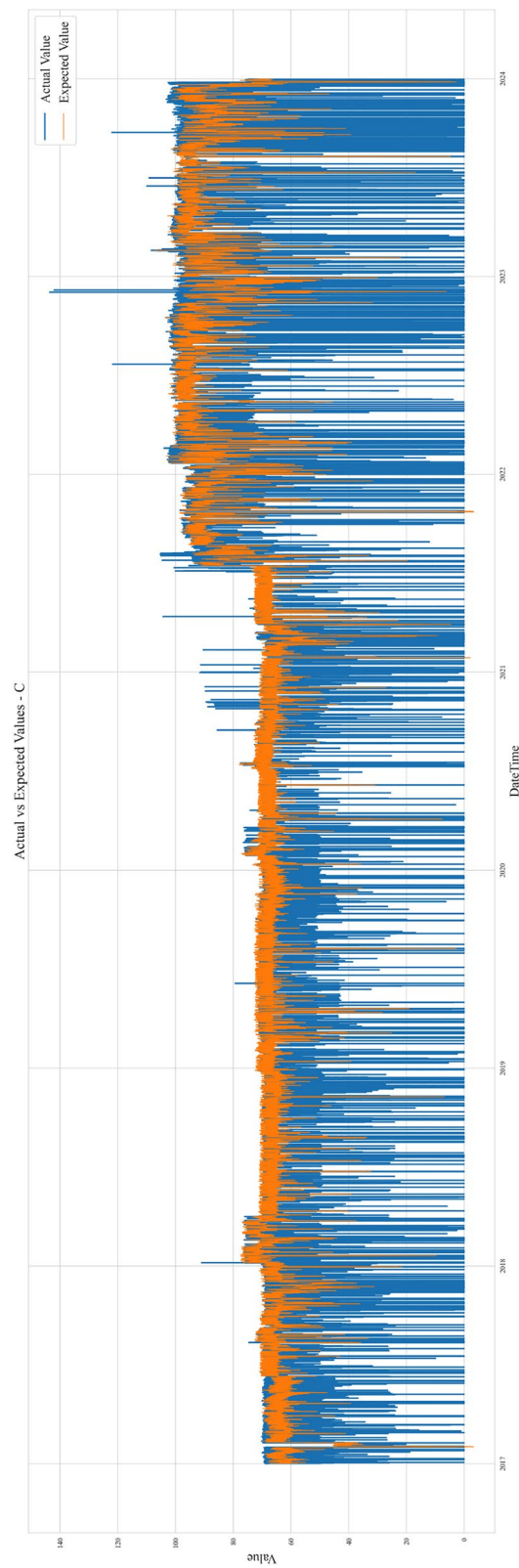


Fig. 15 Analysis of value deviation in Company C

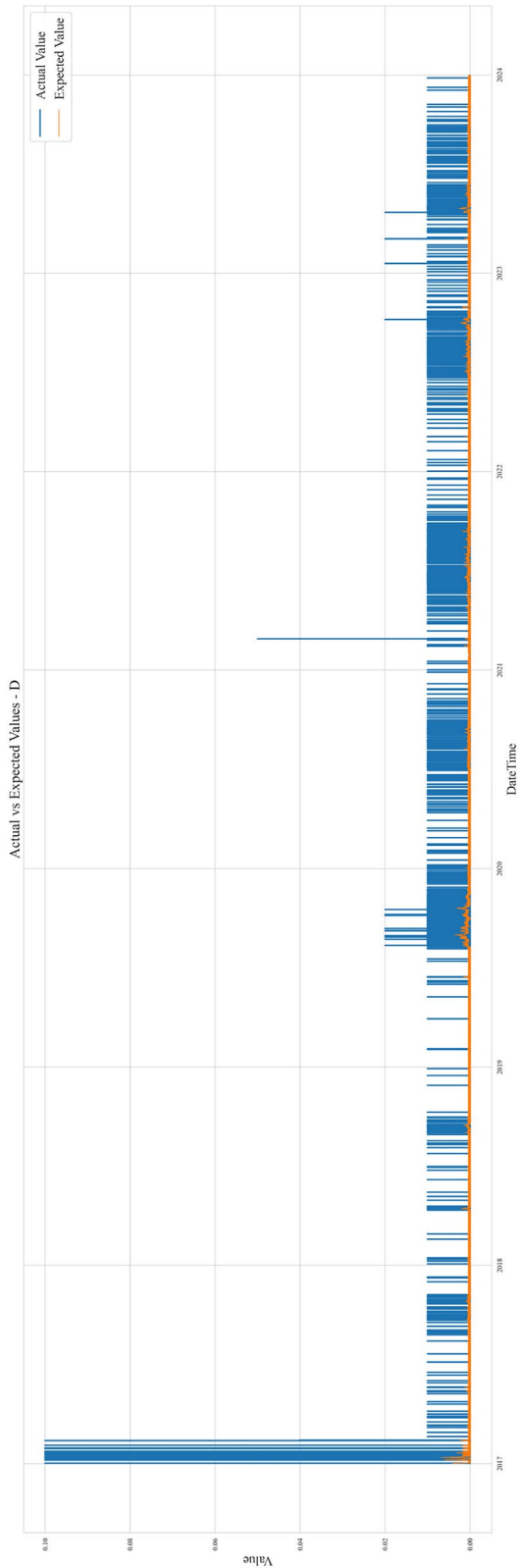


Fig. 16 Analysis of value deviation in Company D

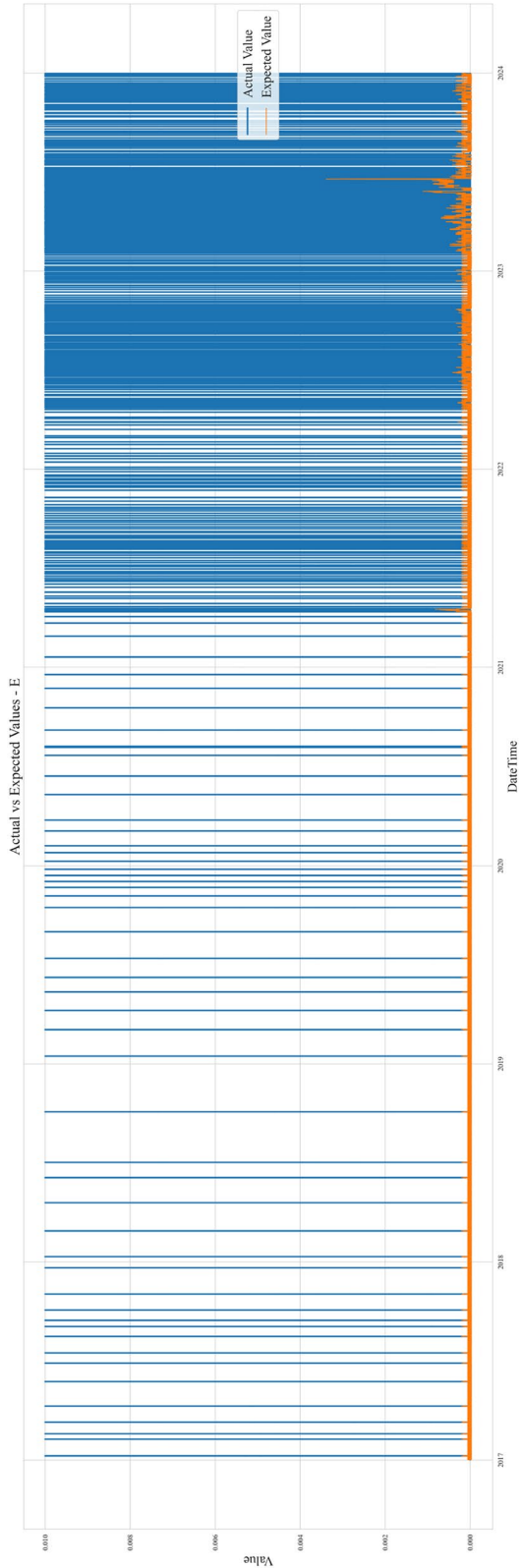


Fig. 17 Analysis of value deviation in Company E

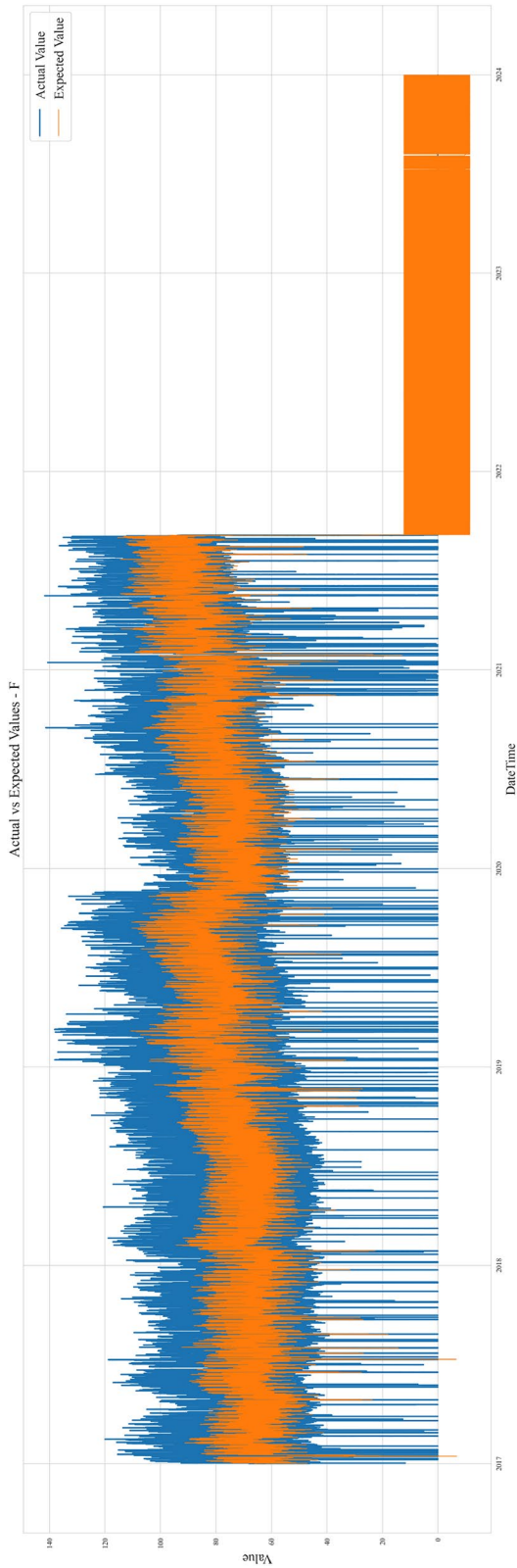


Fig. 18 Analysis of value deviation in Company F

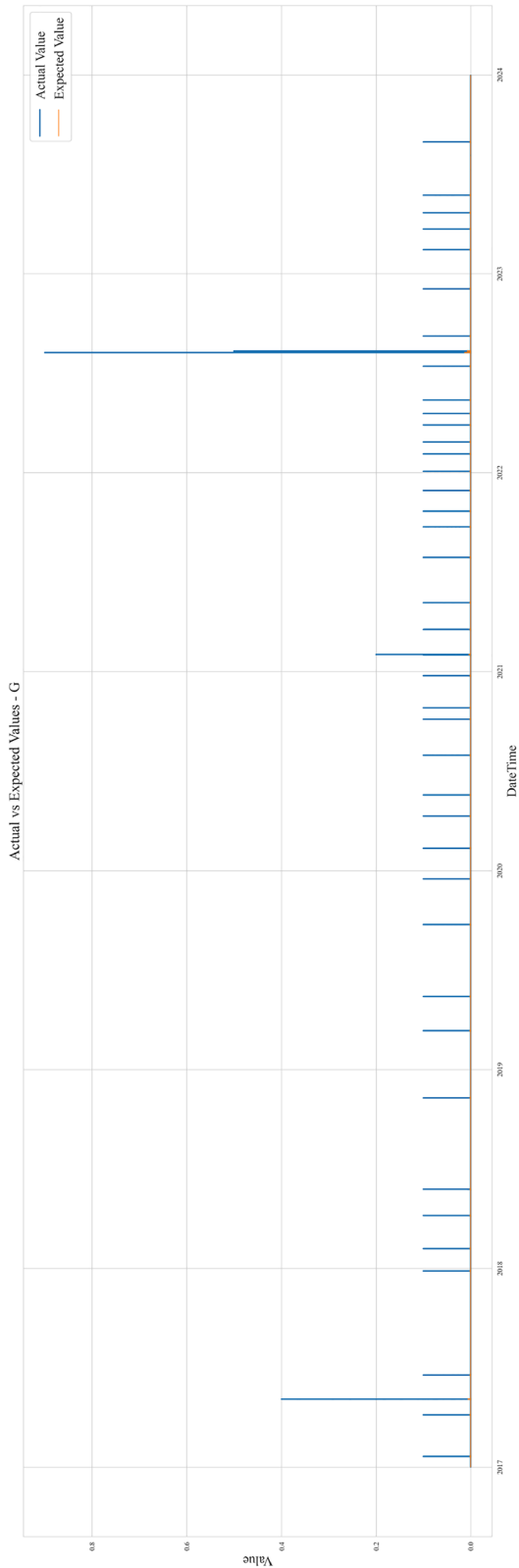


Fig. 19 Analysis of value deviation in Company G

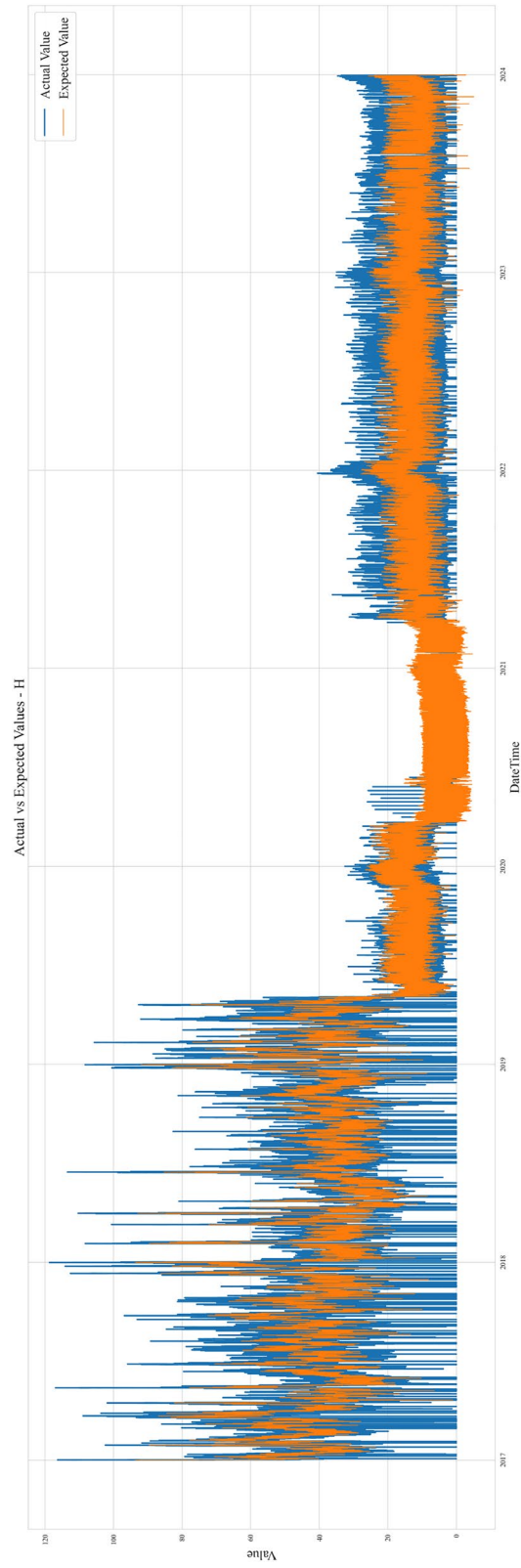


Fig. 20 Analysis of value deviation in Company H

fragmented data and analyzing variations, as detailed in our abstract and methodology sections, represent our efforts to mitigate these inherent challenges and provide a robust analysis of energy usage patterns. Even modern approaches like machine learning and deep learning methods have limitations, both are unable to extract valid information between discontinuous data. Deep learning models like FCNNs suffer from many parameters and overfitting, struggle with high-dimensional data, while CNNs are structurally complex, computationally intensive, and less effective for time series data, and traditional RNNs are challenged by gradient issues in long sequences of data [37].

Despite the availability of historical smart meter data, power load forecasting methodologies face several inherent limitations when dealing with the complexity and strong nonlinear fluctuations characteristic of electricity consumption patterns. Traditional time series analysis methods, while capable of considering trend and seasonality, exhibit a limited ability to model the intricate nonlinear relationships present in energy demand [37]. Similarly, statistical methods often struggle with these nonlinearities and may not effectively account for the impact of external factors like weather variations or holidays [37].

While the historical depth of data, such as the smart meter data spanning 2017 to 2023, provides a valuable foundation for identifying consumption trends, the inherent complexity and volatility of load data present limits for straightforward analysis [38]. Although basic statistical methods can assess the significance of trends, effectively capturing long-term dependencies requires advanced hybrid approaches [38]. These methods, which integrate data smoothing (like Gaussian smoothing), decomposition techniques (like CEEMDAN), and attention mechanisms with deep neural models (like Bi-GRU), are essential for learning from historical data by breaking down complex, nonlinear patterns into more manageable components and focusing on relevant portions of the time series [38]. This is in contrast to traditional forecasting techniques such as ARIMA [13] and exponential smoothing, which often struggle with the significant nonlinearities and abrupt fluctuations inherent in energy consumption data [37, 38].

Even with the inherent limitations of data handling and the struggles of both traditional and modern machine/deep learning approaches in fully capturing the complexities of electricity consumption data, undertaking time series studies remains crucial because they provide a foundational understanding of consumption patterns and trends that are essential for effective energy management, demand response, and long-term planning. Furthermore, the ability to establish consumption trends and capture variations over time using longitudinal smart meter data, despite some limits, is fundamental for assessing the impact of various factors on energy demand and for building more accurate forecasting models in the future

Conclusions

This comprehensive study has pioneered the analysis of energy consumption patterns among Tanzania's large power users through advanced time series techniques applied to TANESCO smart meter data. By addressing critical challenges unique to Tanzania's energy landscape including pervasive data fragmentation due to power interruptions, inconsistent data availability across years, and significant intercompany consumption disparities established a robust methodological framework for transforming

discontinuous datasets into actionable insights. Our approach integrated structural time series decomposition to dissect consumption into trend, seasonal, and residual components, revealing that industrial energy demand is driven by operational schedules (e.g., weekday peaks), seasonal factors (e.g., dry season surges in mining/cooling), and sector-specific growth trajectories. Clustering analysis further segmented eight anonymized companies into distinct behavioral groups, highlighting that high-consumption entities (e.g., Companies C and F) exhibit volatile usage with frequent outliers, while others (e.g., Companies B, D, E, and G) maintain stable, low-consumption profiles. These patterns expose inefficiencies such as off-peak energy spikes and irregular consumption, signaling opportunities for demand-side management. Our findings demonstrate that temporal granularity in smart meter data even at 20-minute intervals enables utilities to optimize grid management by anticipating load fluctuations linked to industrial cycles, and target energy-saving interventions (e.g., shifting nonessential operations to low-demand periods). For Tanzania, these insights translate to tangible economic and operational benefits: reduced strain on aging infrastructure, cost-savings through peak shaving, and improved integration of renewable energy sources. Methodologically, our work underscores the viability of time series analysis in resource-constrained environments, proving that fragmented data can yield reliable trends when processed through rigorous imputation and standardization protocols. While limitations around data volatility persist, this study lays a foundational framework for evidence-based energy governance in Tanzania and similar emerging economies.

Future work to combat the limitations could focus on several key areas: To address missing data caused by power interruptions, more advanced imputation techniques beyond forward-filling, such as machine learning methods, could be explored, along with directly modeling the impact of these interruptions. For high data fluctuation and variability, incorporating external factors like weather and economic indicators and employing robust statistical methods for noise reduction could be beneficial. Sudden unpredictable changes in energy use could be handled by integrating anomaly detection and investigating their underlying causes. To refine the data, sophisticated feature engineering and selection, as well as developing automated methods for detecting and correcting data inconsistencies, are recommended. Finally, applying advanced deep learning techniques for time series forecasting may inherently improve the ability to model complex patterns.

Abbreviations

$\ x - \mu_i\ $	Euclidean distance between data point x and centroid μ_i
C_i	The i -th cluster
J	Within-cluster sum of squares (WCSS)
k	Number of clusters
n	Number of observations (data points)
$R(t)$	Residual component of energy consumption at time t
$S(t)$	Seasonal component of energy consumption at time t
$T(t)$	Trend component of energy consumption at time t
x	Data point (feature vector) in clustering context
$X(t)$	Observed energy consumption at time t
μ_i	Centroid of the i -th cluster
μ_R	Mean of the residuals
$\sigma_R(t)$	Standard deviation of the residuals at time t
τ	Threshold parameter for outlier detection

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Author Contributions

DR provided the concept and the objectives regarding the work. EN worked on the related and the methodology sections. EY performed the paper structuring, results, and discussion part. NM worked on data visualization and interpretation. All authors contributed in writing all sections of the paper, read the article, and approved it before submission.

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Data Availability

The data used for this research is available at the site provided with permission from TANESCO.

Declarations

Conflict of interest

The authors declare no Conflict of interest related to this research. Data provided by TANESCO has been used solely for academic and research purposes, ensuring confidentiality and compliance with ethical standards.

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